Available ECG sensors:

AD8232:

AD8232 is a commonly used ECG sensor module for monitoring heart activity. It is widely used in student projects and biomedical applications because it is easy to work with and available at a low cost.

One advantage of AD8232 is that it can be easily connected with platforms such as Arduino and ESP32, making it suitable for testing and development.

However it requires external electrodes for signal acquisition, and the ECG signal may be affected by body movements.

For our project

AD8232 can be used to obtain the ECG signal required for Pulse Transit Time (PTT) calculation. Since it is affordable, readily available, and simple to integrate, it is a suitable option for the initial prototype and it is cost effective

MAX30003

MAX30003 is an ECG sensor IC mainly developed for wearable health monitoring devices. It is capable of providing good-quality ECG signals while consuming very little power, which is important for battery-operated devices.

Compared to AD8232, it is more suitable for wearable applications because of its compact size and low power requirements.

However it is not as easily available in the local market and usually requires a custom PCB for implementation. In our project, MAX30003 can be used to acquire the ECG signal needed for PTT calculation.

ADS1292R

ADS1292R is a high-performance ECG acquisition IC used in medical and research applications. It is known for its high accuracy and ability to capture ECG signals with good quality. Because of its advanced features, it is often used in systems where reliable ECG measurements are important. However, the hardware design is more complex and the cost is higher compared to AD8232.

For our project, ADS1292R can be considered if very accurate ECG data is required. While it may not be necessary for the initial prototype, it can be useful for advanced development and detailed signal analysis.

Available ppg sensors:

MAX30102

MAX30102 is a widely used optical sensor for monitoring heart rate and SpO_2 levels. It is commonly seen in wearable projects because of its low cost and easy availability. The sensor is simple to connect with microcontrollers and can provide the PPG signal needed for pulse detection. One drawback is that the readings can be affected by hand movements or improper sensor placement. MAX30102 is a practical option for testing and developing PTT-based blood pressure monitoring systems.

MAX86141

Unlike MAX30102, MAX86141 is designed for more advanced wearable health monitoring applications. It offers better signal quality and performs more reliably when the user is moving. This makes it suitable for continuous monitoring where stable PPG signals are important. The sensor is more expensive and may require additional effort during hardware development, but it provides better overall performance. For a wearable blood pressure monitoring device, MAX86141 can be a strong choice when accuracy and signal quality are given higher priority.

AD8232 + MAX30102

This combination is suitable for prototype development. Both sensors are low cost, easily available, and simple to use. It is a good choice for testing the PTT algorithm and validating the project concept.(good for prototyping cost effective)

Combinations that can be used:

MAX30003 + MAX30102

This combination provides better ECG signal quality while keeping the overall cost reasonable. It can be used when a more wearable-friendly ECG solution is needed.

MAX30003 + MAX86141

This combination is suitable for wearable applications. Both sensors provide good signal quality and lower power consumption. It is a good option for developing the final version of the device.

ADS1292R + MAX86141

This combination offers the best signal quality among all the options. However, it is more expensive and requires a more complex hardware design.